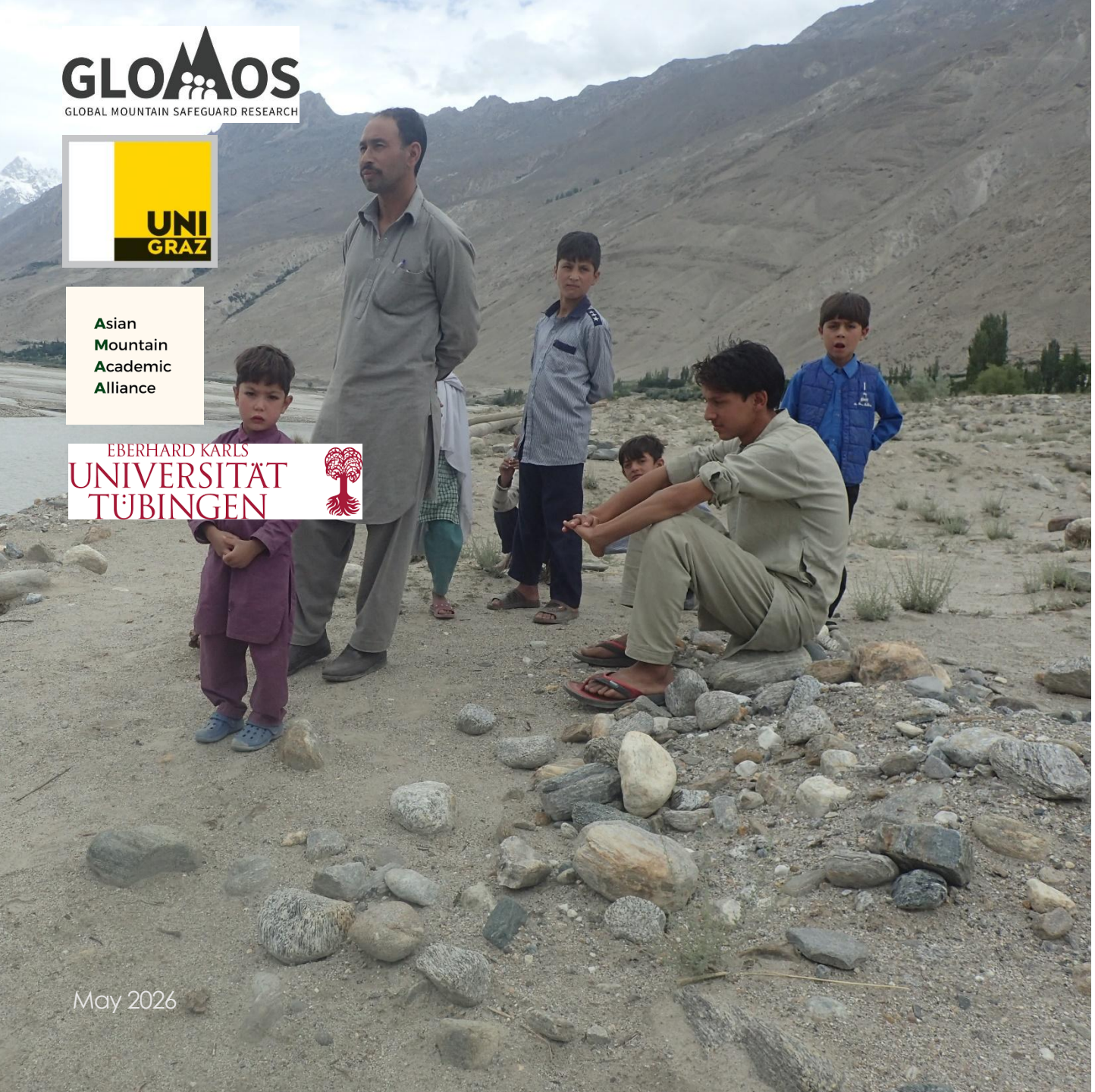


climate risk in vulnerable mountain regions of the world

Day 4 (29 May)

Jakob Steiner

jakob.steiner@uni-graz.at

fidelsteiner.github.io

Course content

22 May (1400 – 1700)

Pre course evaluation - general introduction – the mathematics of risk – introduction to assignments

27 May (0900 – 1500)

Examples of hazards / exposure types /vulnerability assessments

Communicating risk

28 May (0900 – 1500)

International initiatives

Science to policy

29 May (0900 – 1500)

Serious game – course wrap up

Course content

29 May (0900 – 1500)

09:00 – 10:00	new data and methods
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The future of early warning

Impact based forecasting (https://www.anticipation-hub.org/learn/emerging-topics/impact-based-forecasting)

- ‘What will the weather do’ rather than ‘What will the weather be’
- What is useful for people?

Government of Pakistan
Ministry of Defence (Defence Division)
Pakistan Meteorological Department
Research and Development Division
Islamabad, Pakistan

GLOF ALERT

GLOF/Met-3(2)/2025/

Dated: 02nd April 2026
Issue Time: 03:00 PM

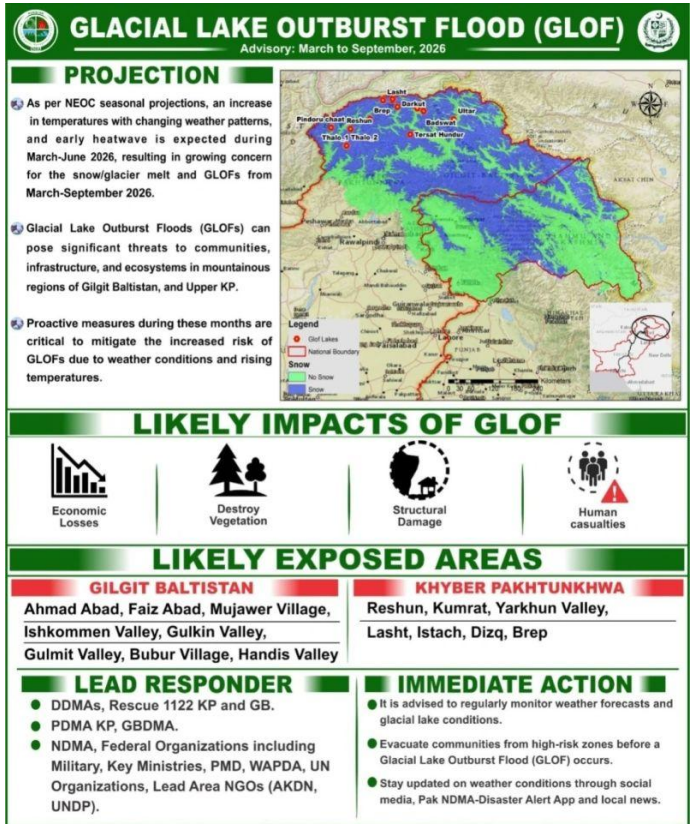
A westerly wave is likely to affect upper parts of the country from the night of 02nd April and is expected to persist till 08th April with intermittent rain. Under the influence of this weather system, widespread rain/thunderstorm and gust, accompanied by moderate to heavy rainfall and snowfall over hill tops, is expected in Mankiyal, Matiltan, Kumrat and Utror-gabral valleys of Swat, and Kandia valley of lower Khyber Pakhtunkhwa (KPK) and Resun-Booni, Arkari and Madaklasht of Chitral upper KPK. The districts of Gilgit Baltistan (GB) Ghizer, Hunza, Nagar, Astore, Guanche, Diamer, Kharmong, Skardu and Shigar are likely to be affected. The expected persistent precipitation, coupled with the snowmelt in high-altitude catchments, and intense runoff generation may significantly aggravate hydro meteorological hazards in vulnerable valleys.

Likely Weather-Induced Impacts:

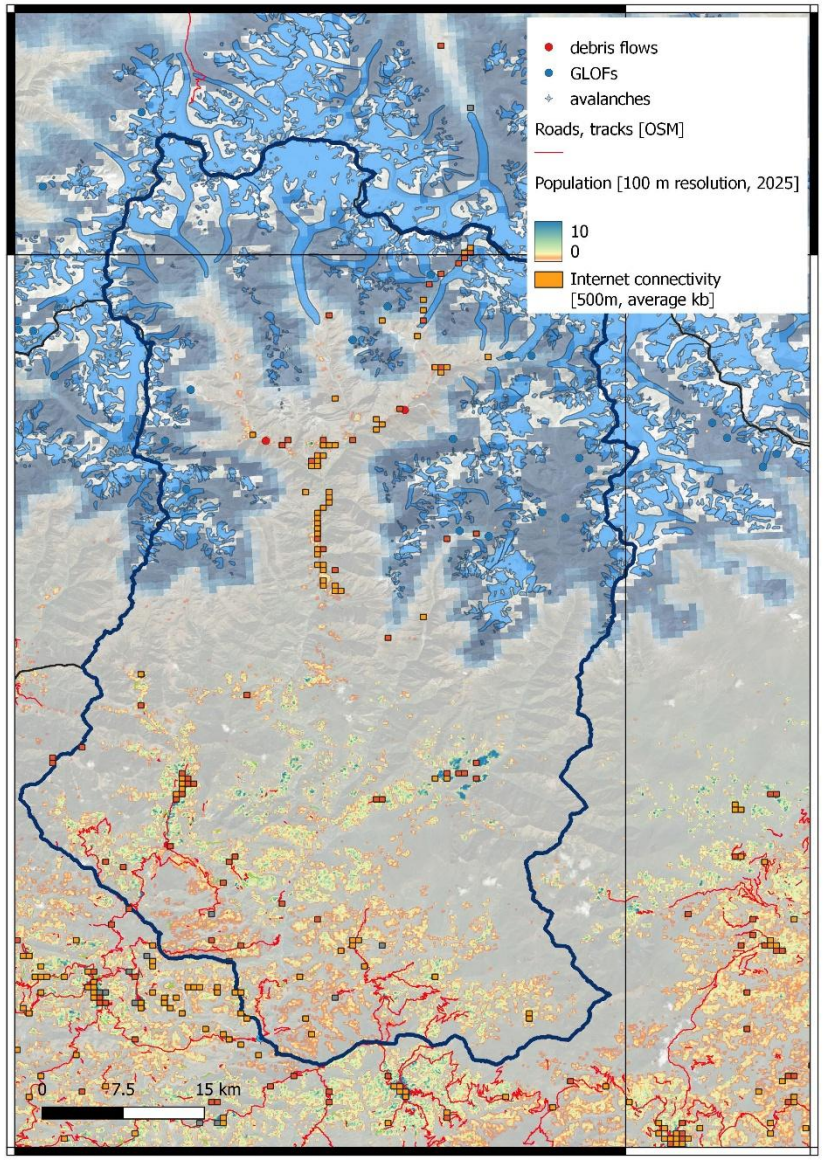
- Probability of Glacial Lake Outburst Flood (GLOF) in vulnerable glaciated basins and downstream settlements.
- Flash flooding is likely in local streams, nullahs and tributaries particularly in KPK river basins.
- Heavy rainfall over steep slopes may trigger landslides, mudslides, debris flow in mountainous districts of KP and GB.
- Mountain roads and communication links may face temporary closure/blockage due to debris flow and flash runoff.

2. All concerned are advised to remain alert and take necessary measures to avoid any untoward situation.

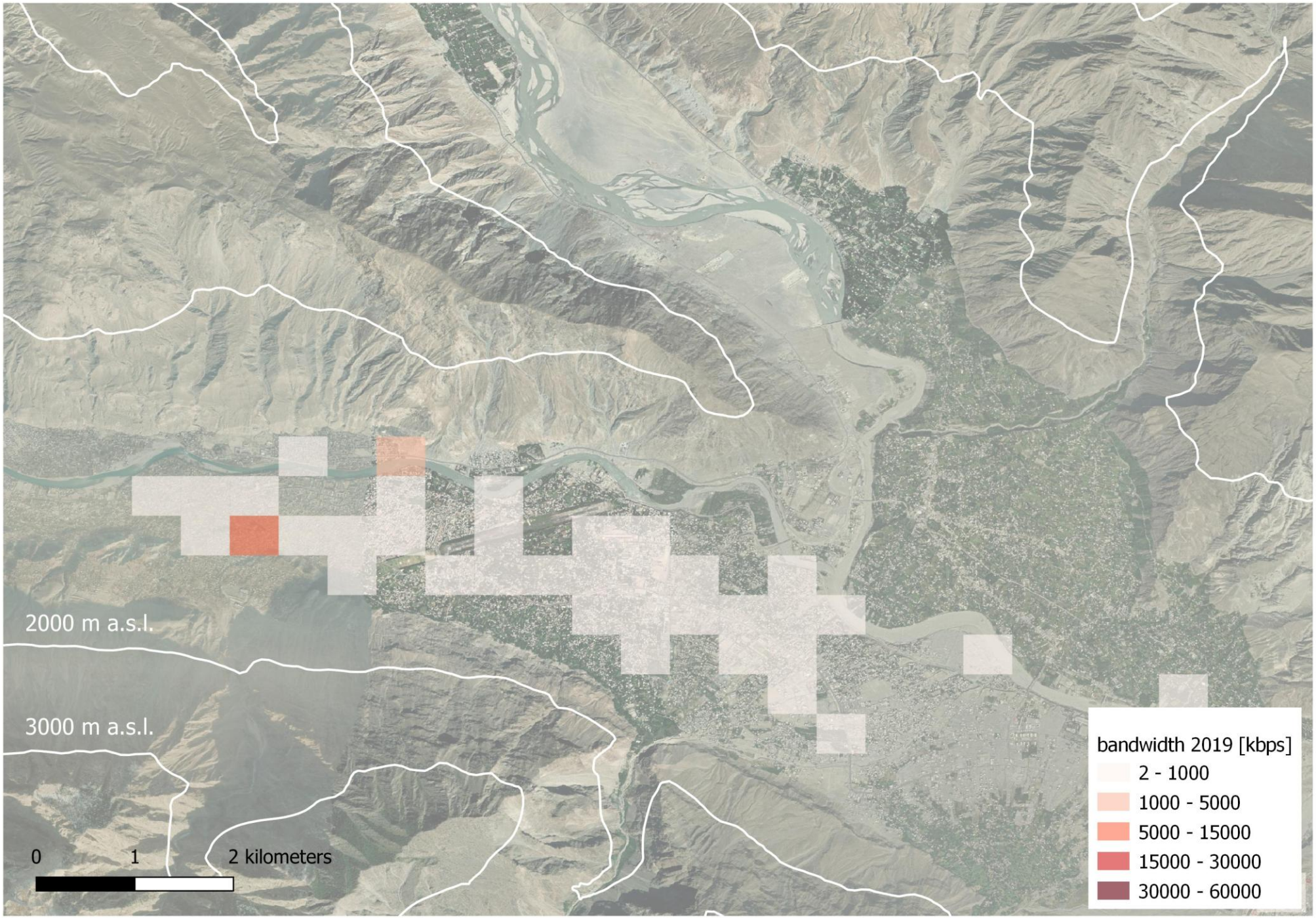
(Zeenat Yasmeen)
Meteorologist,
For Director General, PMD



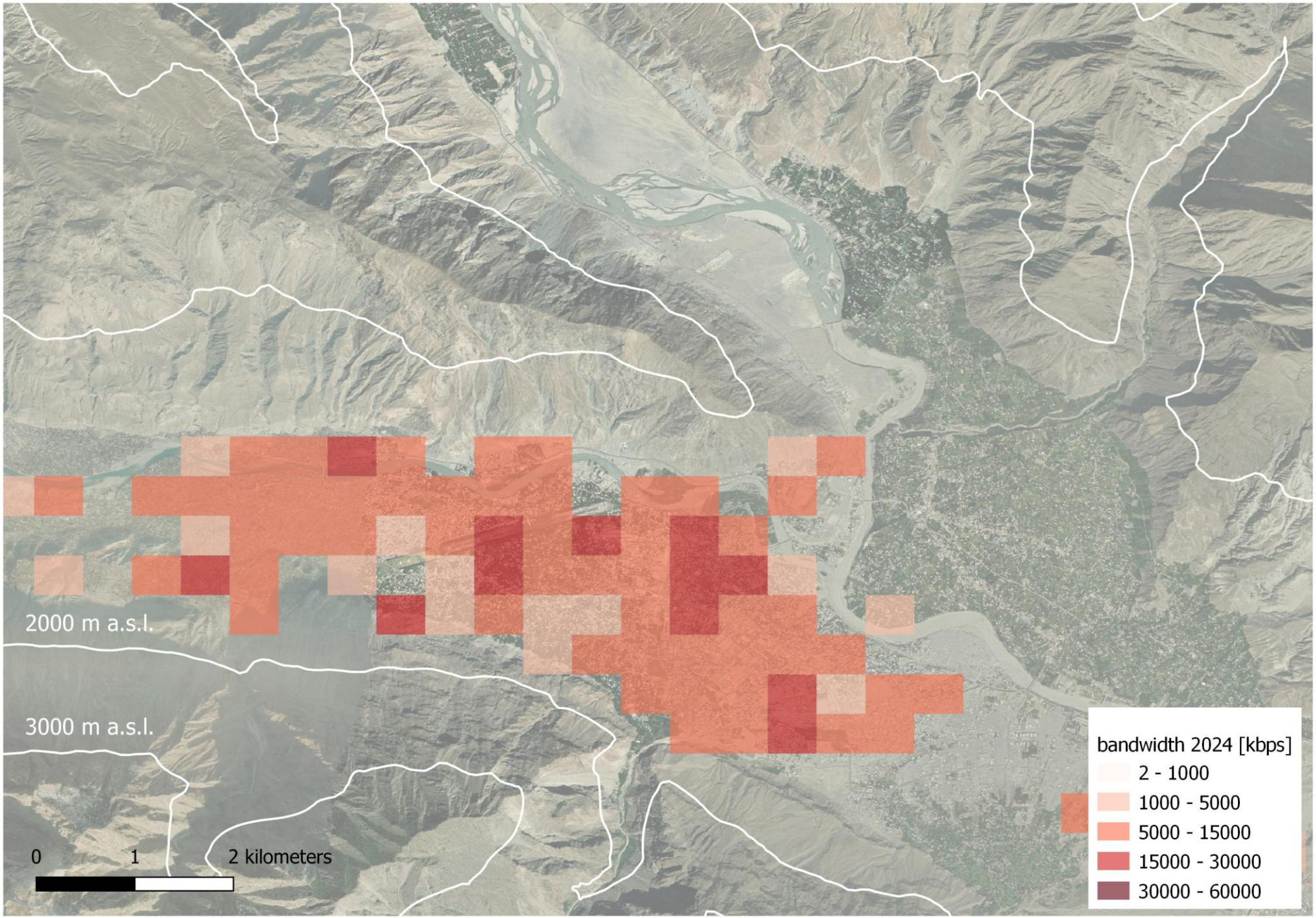
The future of tracking data access



Ookla data



Internet connectivity
Ookla 2019



Internet connectivity
Ookla 2024

Citizen Science

Integrating Citizen Science and Data-Driven Approaches for Hydrological Hazard Assessment in the Kathmandu Valley

PROJECT PROPOSAL

Implementing Organization:	Smartphones For Water Nepal (S4W-Nepal)
Supporting Organization:	SmartPhones4Water
Geographic Focus:	Kathmandu Valley, Bagmati Province, Nepal
Focal Watersheds:	Bagmati River Basin, Nakkhu & Hanumante Sub-watersheds
Project Duration:	12 Months (2026–2027)
Submitted:	May 2026

https://s4w-nepal.smartphones4water.org/





From conventional to the future of forecasting

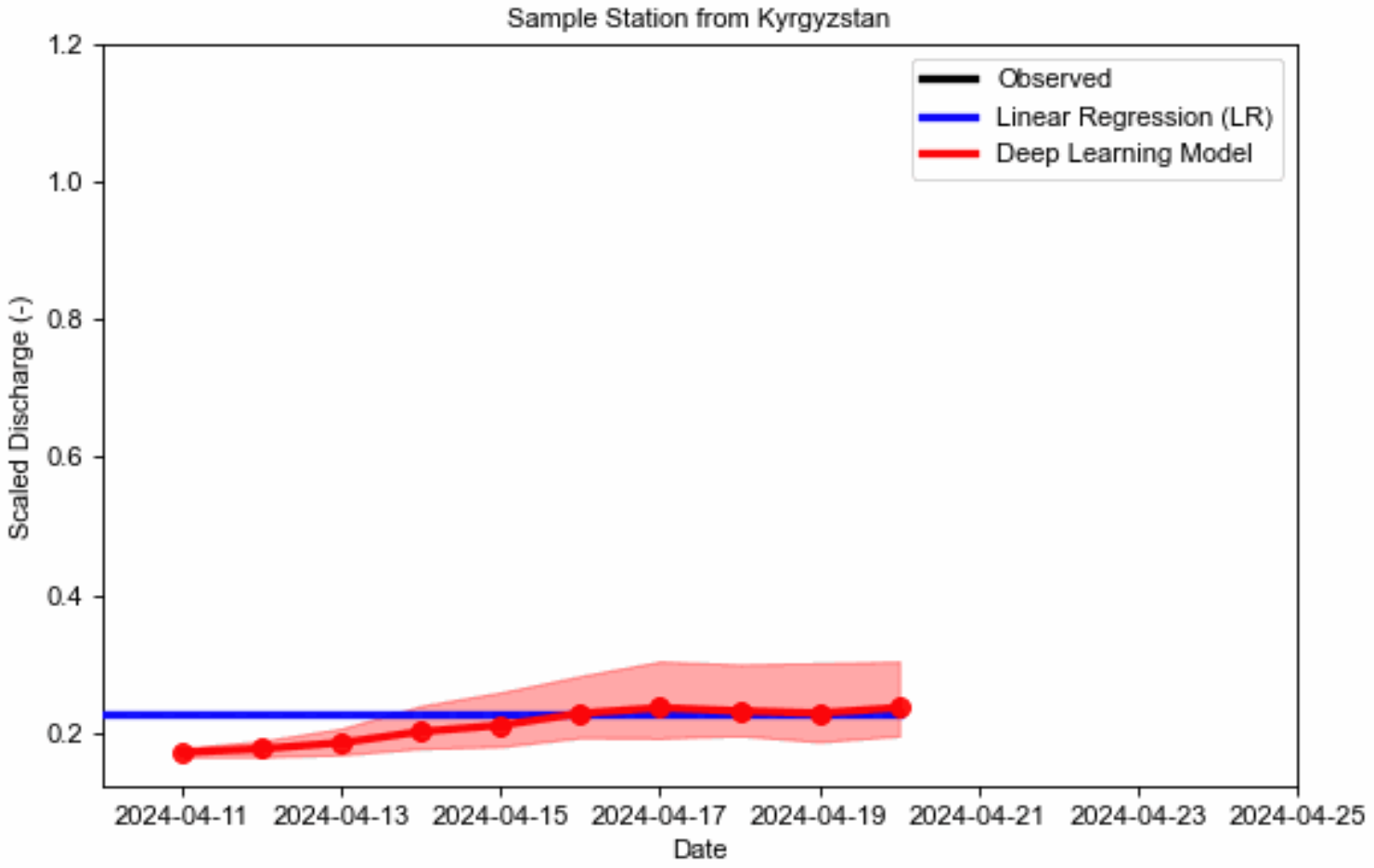


Figure 1: Traditional water level measurements are carried out manually in Central Asia twice per day. These traditional workflows are gradually replaced by modern montioring infrastructure that measures many relevant variables at high-frequency. These creates a data flood overwhelming the local institutions. Source: hydrosolutions GmbH.

How do we bring conventional ways of monitoring into the age of Big Data, ML and AI?

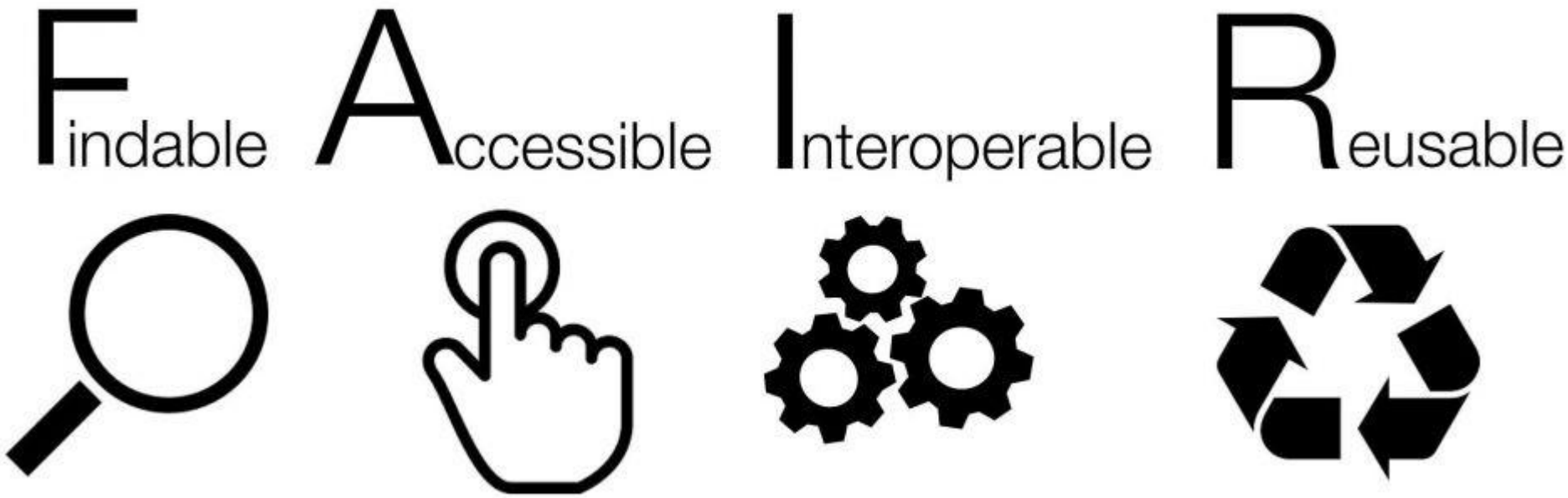
https://www.hydrosolutions.ch/

From conventional to the future of forecasting



Is FAIR data enough?

Is FAIR data enough?

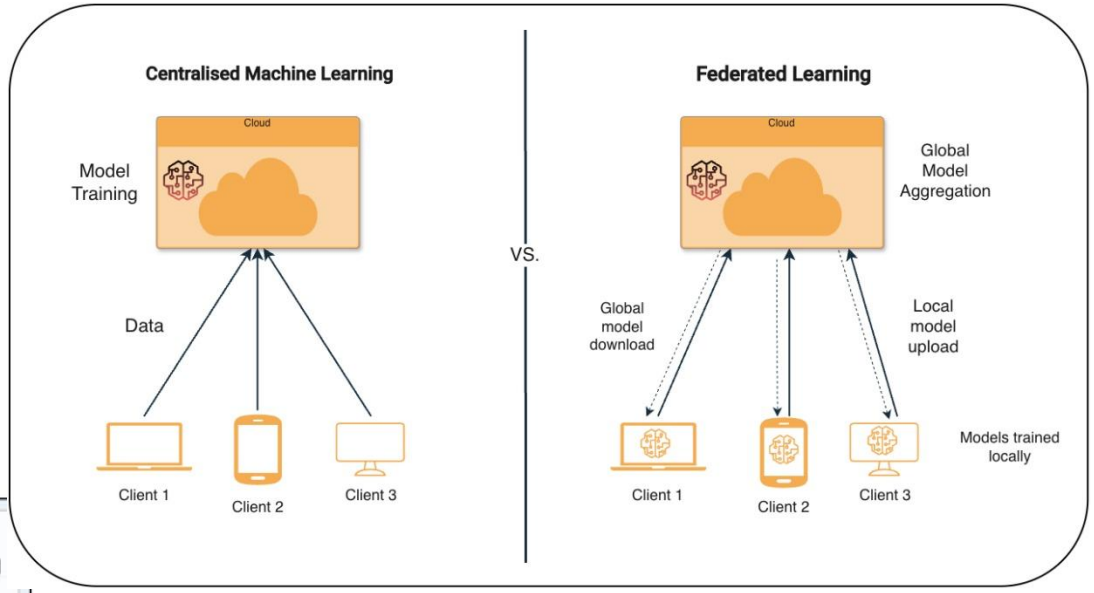


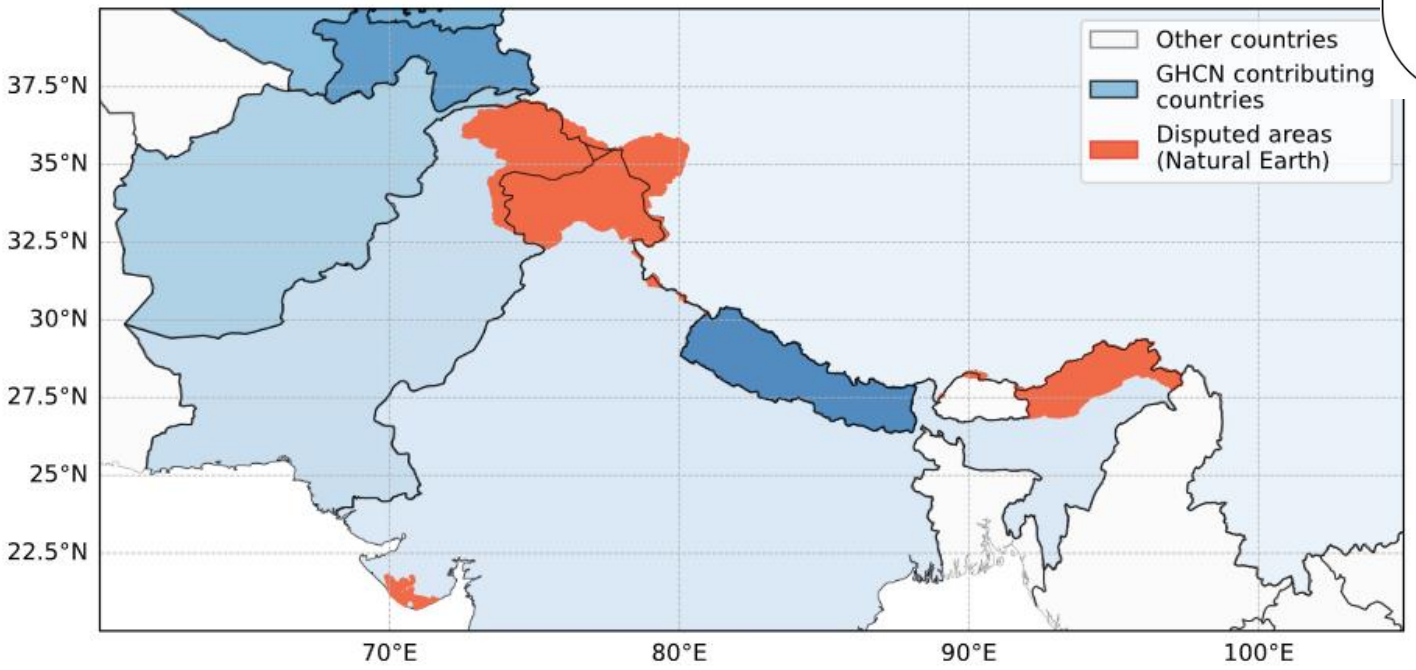
CARE data



https://www.gida-global.org/care-principles-copy

Federated Learning – how to model when countries don't share?



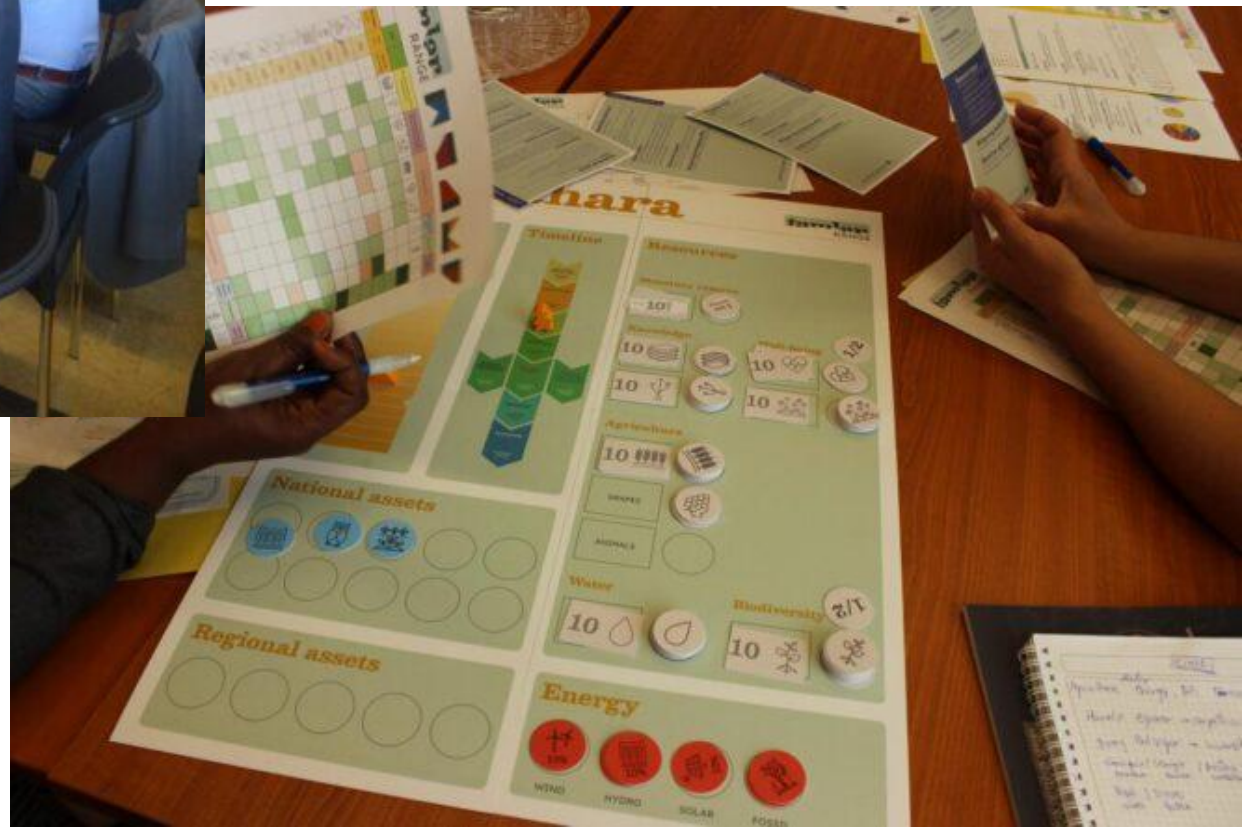


Bilateral deliberations



https://mountains-connect.org/interregional-workshop/




National assets
Regional assets
Water
Energy
10
WIND
HYDRO
SOLAR

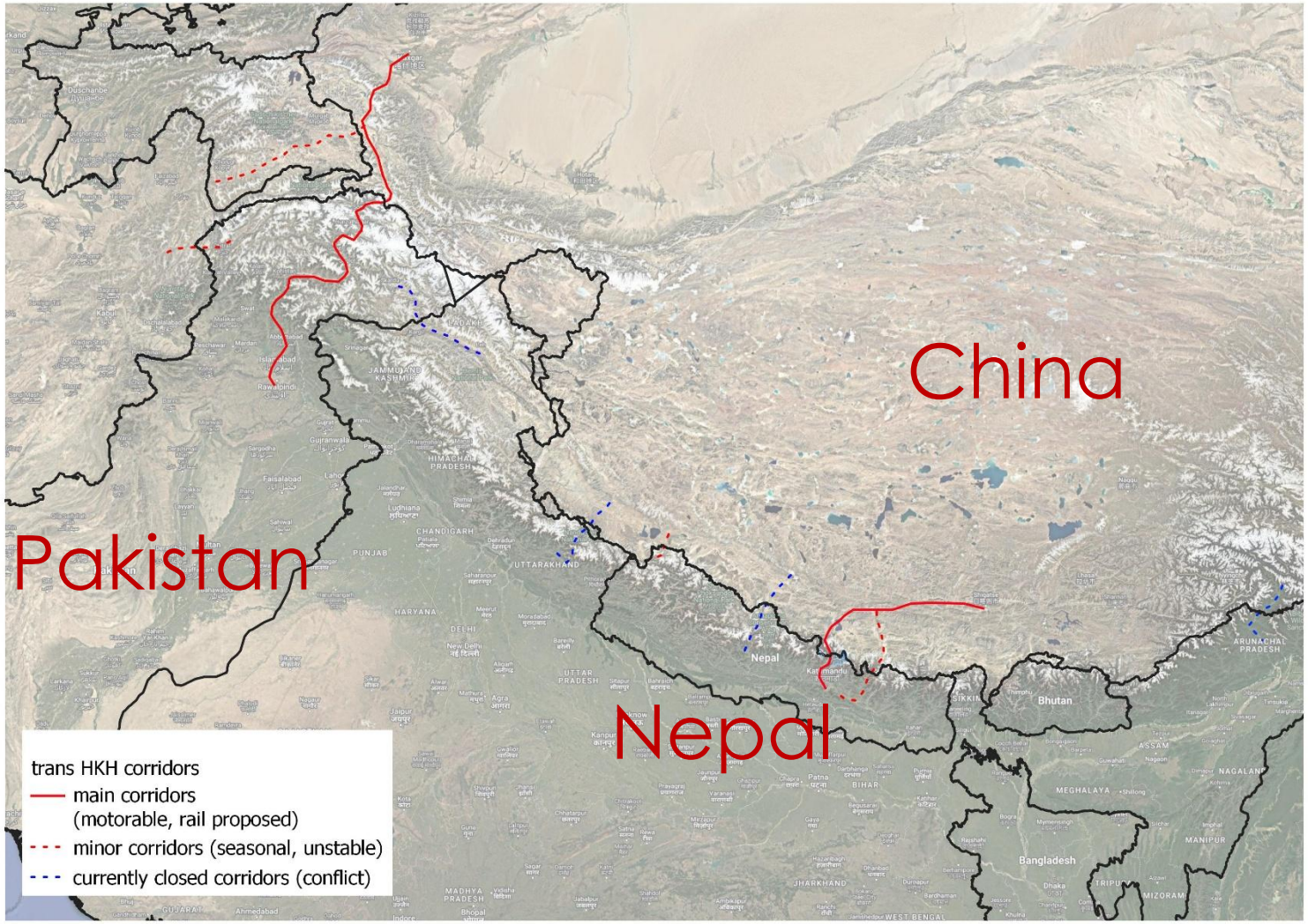

UN

Bilateral Deliberations

China, Pakistan and Nepal are three countries that have generally favorable relations and who engage diplomatically, on trade, education and culture.

They come together to discuss on how to address mountain risks in their border regions but possibly also at a larger scale.

Can they agree on a way ahead?



Bilateral Deliberations

A	B	C
	Role	Student Name
China	Ministry	
	Business (cars)	
	Business (construction)	
	Academia	Ekaterina Fedoseeva
Nepal	Civil Society	Laura Reutter
	Minister	
	Academia (Fores	Valerie Kalt
	Academia (Clima	Rachel Lecroix
	International Age	Xuan Han Chen
Pakistan	Civil Society	
	Business	Sebastian Buțu
	Ministry	
	Academia	David Fulks
	International Organiation	

1) Find common ground in your countries

What are your priorities? What are your negotiation strategies?

2) Find common ground in focus groups:

(a) Academic

(b) International Organizations/Civil Society/Business

With knowledge from your country deliberations, can you agree on priorities with colleagues?

3) Plenary:

(a) Present and defend your proposals

(b) Find and present a common agreement

To consider while you negotiate.

Try to be realistic, not just smart – but also, take the chance and be progressive, innovative and creative – what do you think would be the best agreed path forward to achieve progress? How would that be communicated? To whom?

Consider current global geopolitics. How do they influence your deliberations?

Don’t be irritated by the lack of time – that’s how it goes in *real life* too.

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China – Nepal – Pakistan Summit on transboundary mountain risks



Plenary Proceedings

Tabled motions:

Academia: sharing of the Himalayan climate – joint effort in research, exchanging databases, research findings, joint glacier research, Nepal can share the mountaineering experience. Shared monsoon forecasting. Scholarship exchange. Offer research infrastructure. If funds not available in the region, India could fund – Nepal has good history. Settle on collecting data on common topics, ask China to bring in funding. China can provide scholarships, collaborate on research centers.

Non-academic stakeholders: Collaboration or joint operation in the hydropower domain – diminish effects on local populations – establish working group to establish guidelines. Funding ideally from ADB, if restrictions lenient, otherwise funding would need to come from the EU/US, use this as a chip in trade negotiations. Reduce dependence on Chinese technology. Many Indigenous groups in Nepal, require risk assessments keeping their presence/ownership in mind. Exchange data with China on people across borders. Exchange data with relation to hydropower operations – find funding as a group. Bank focuses on regional rather than single country investment – cooperation encouraged. Funding needs to be transparently used. Reporting on risk evaluation required. Locals need to benefit.

Plenary proceedings (II)

Nepal requires support on academic institutions – student exchange favored with all countries. Exchange of satellite data. Advance in terms of development.

Exchange of databases – academics provide data that is required in the hydropower domain. Academics provide reports. Develop mountain risk protocol, to be understood by non academics. Non academics transmit it to the general public. Impact assessment for infrastructure development. Both academics and non-academics need to play part in a dialogue.

Concrete steps:

Start sharing data.

Get early warning mechanism in place (neighboring countries need to come in – learn from countries where EWS work already).

Plenary proceedings (III)

China building a big road in Nepal – Nepal concerned with impacts, impact assessments need to be rigorous

Addressing overreliance on Chinese infrastructure and funding.

China can provide protection of strategic infrastructure corridors.

Chinese Professors coming to Pakistan/Nepal to teach. Export know how.

Negotiations are officially over – you're asked by your media to report success or failure

Did you achieve what you wanted to get through? Where did you need to compromise?

Is there self criticism for your own country, your own interest group?

Did you see potential to find global common ground for interests of mountain regions?

How would you change your strategy for next time?

What stakeholder need to be brought in for next deliberations?

Negotiations are officially over – you're back to your Self

Negotiations are officially over – you're back to your Self

Could you find agreements with others? What made coming to agreements or strategies most difficult? What were you lacking to come to agreements?

Is virtual exchange useful? Do you need to see the others in person, do you need to travel to their countries?

Do you think it is realistic to come to agreements in such a setting and what would it take that they are being adhered to?

We're done … time for a few more tasks

Course self evaluation (Google Form, Moodle)

University Tübingen course evaluation

Don't forget to upload the writing assignment on Moodle (under 1st Day)!

We’re done … time for a look back

Your final thoughts on risks in mountain regions?

Your questions?